

Lab 1: Blockchain and Transaction

Objectives:

- To know about different types of blockchain and how their transaction is managed using a crypto wallet. Also, we will discuss some blockchain explorers

Submission:

- Three checkpoints.

Introduction:

In this lab, we will discuss various types of blockchain and how their transactions can be managed from a user's perspective using a cryptocurrency wallet. In addition, we will discuss some blockchain explorers of Ethereum and Bitcoin so that the students can get a real-life idea of what really happens when a transaction is completed.

We know there are primarily two types of blockchain. To interact with a blockchain, a user needs an account, which is identified by an address. To use and manage the accounts, we need a crypto wallet. This wallet primarily stores account information and balance, and also facilitates the easy and efficient management of multiple accounts. To clear your imagination about the crypto wallet and account, you can assume this with a physical wallet(crypto wallet) where you store multiple credit/atm card accounts.

There are primarily two types of crypto wallets such as custodial and non-custodial wallets. In this lab, we will use a non-custodial wallet called Metamask, which can be used as a browser extension.

Section-1 : Setting up Metamask

To install MetaMask,

1. First, open Google Chrome. Click the extension icon at the top-right corner of the browser and then click manage extensions(Figure 1.a). This will bring up the existing extension of your browser. Click the hamburger menu button(menu button with three lines) at the top-left corner of your browser and click Open Chrome Web Store (Figure 1.b & 1.c). The steps are mentioned below:

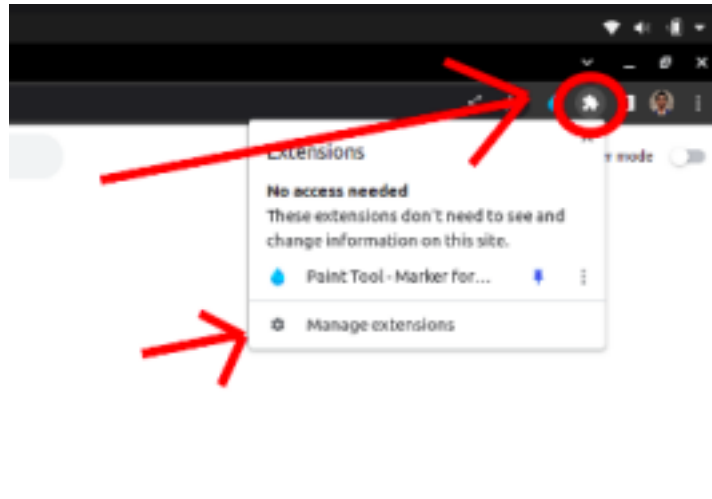


Fig: 1.a

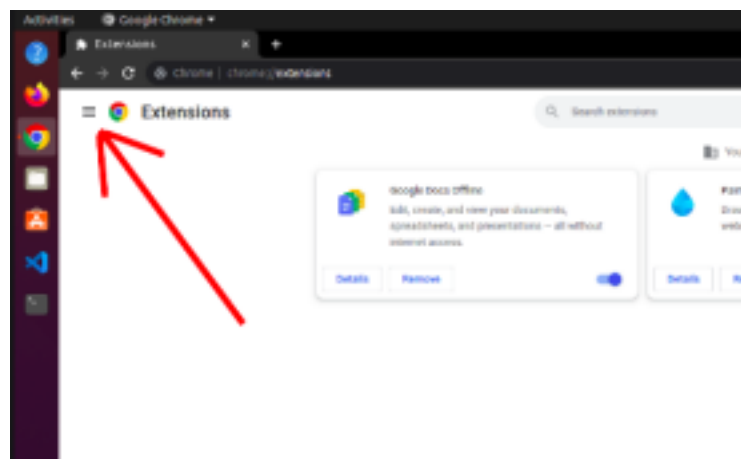
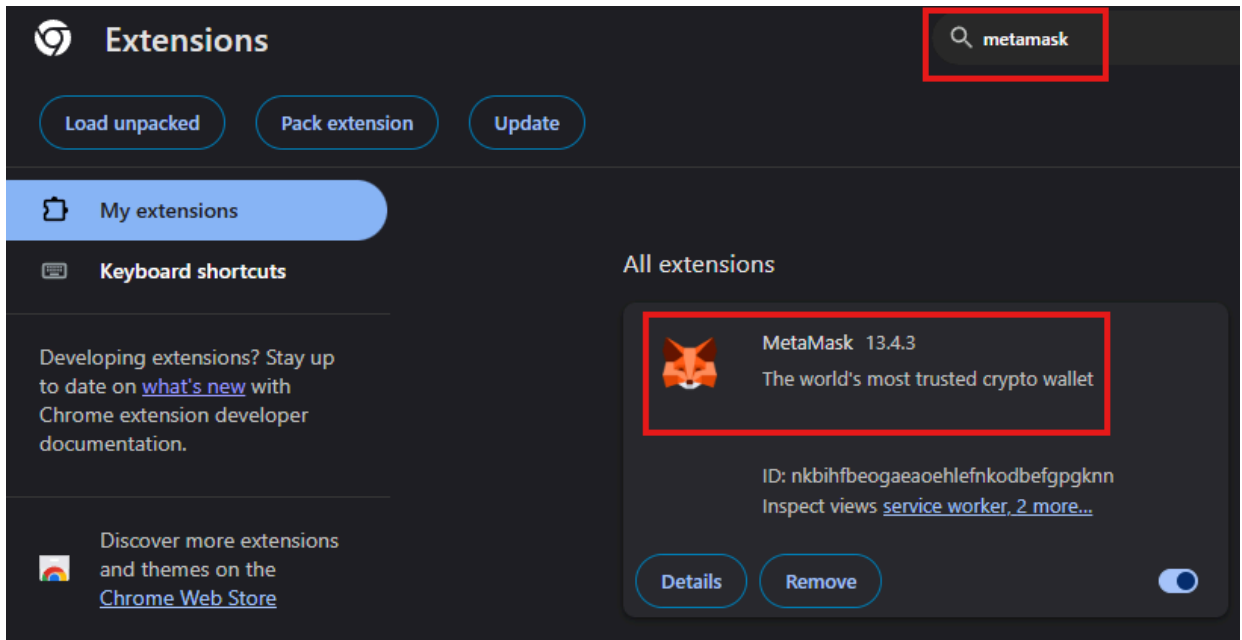


Fig: 1.b

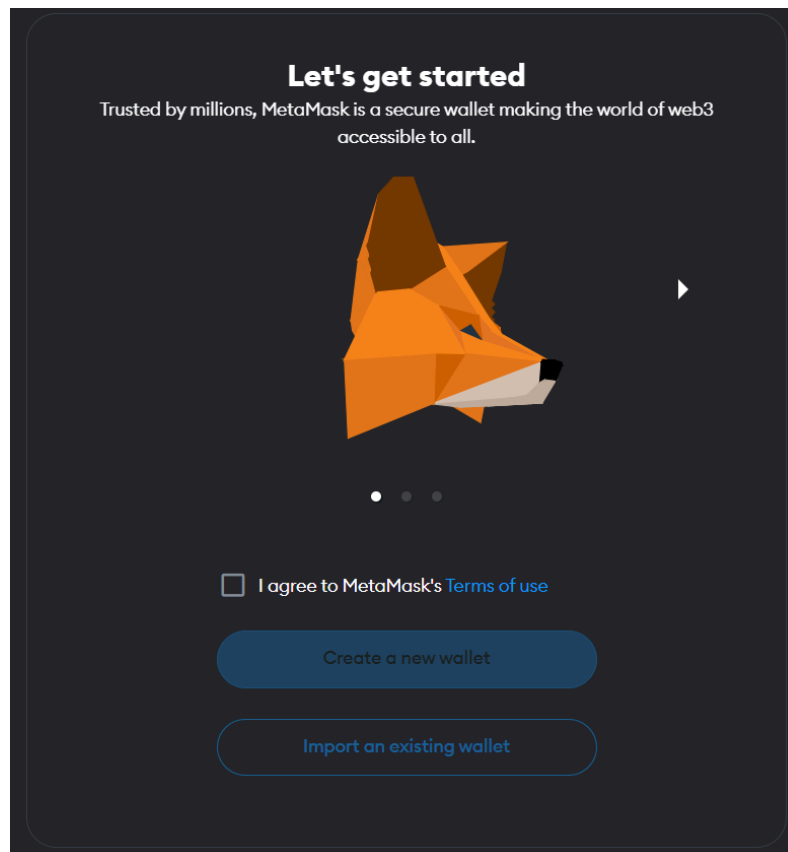


Fig: 1.c

2. Now, search Metamask and click the Metamask wallet shown below:



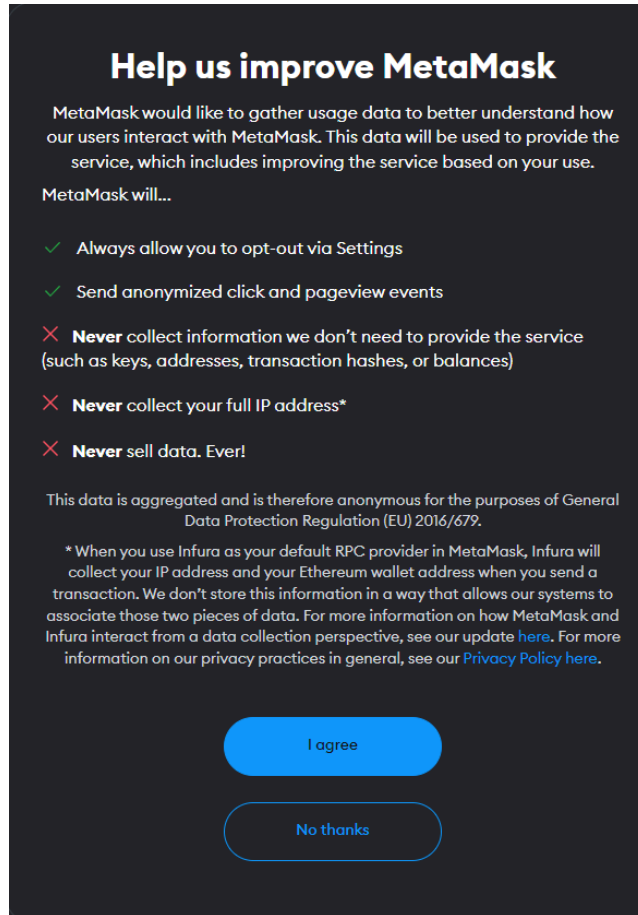
3. Next, click **Add to Chrome**, and this will start downloading and installing the file. After installation, you will see an image like below. We are assuming that you do not have a wallet. Therefore, click the **Create New Wallet** button.



4. A new page will appear. When you have to provide your credentials, and after writing

them, click **Create a new wallet**.

5. A new page will appear. Press **I agree**.



Help us improve MetaMask

MetaMask would like to gather usage data to better understand how our users interact with MetaMask. This data will be used to provide the service, which includes improving the service based on your use.

MetaMask will...

- ✓ Always allow you to opt-out via Settings
- ✓ Send anonymized click and pageview events
- ✗ **Never** collect information we don't need to provide the service (such as keys, addresses, transaction hashes, or balances)
- ✗ **Never** collect your full IP address*
- ✗ **Never** sell data. Ever!

This data is aggregated and is therefore anonymous for the purposes of General Data Protection Regulation (EU) 2016/679.

* When you use Infura as your default RPC provider in MetaMask, Infura will collect your IP address and your Ethereum wallet address when you send a transaction. We don't store this information in a way that allows our systems to associate those two pieces of data. For more information on how MetaMask and Infura interact from a data collection perspective, see our update [here](#). For more information on our privacy practices in general, see our [Privacy Policy](#) [here](#).

I agree

No thanks

6. Now set the password to unlock Metamask and remember it.

Note: in case you forget this password, you have to reset it using a secret phrase (You will learn more about it later).

1

2

3

Create passwordSecure walletConfirm secret recovery phrase

Create password

This password will unlock your MetaMask wallet only on this device. MetaMask can not recover this password.

New password (8 characters min)

Show

Confirm password

☐ I understand that MetaMask cannot recover this password for me. [Learn more](#)

Create a new wallet

7. Afterward, this new page, “Secure your wallet,” will pop up.

1

2

3

Create passwordSecure walletConfirm secret recovery phrase

Secure your wallet

Before getting started, watch this short video to learn about your Secret Recovery Phrase and how to keep your wallet safe.

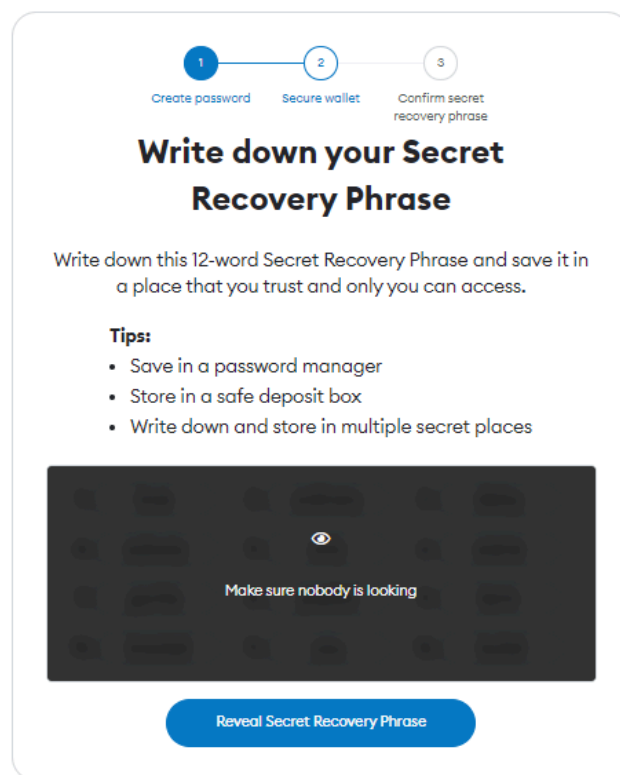
0:00 / 1:35

Remind me later (not recommended)

Secure my wallet (recommended)

8. Click **Secure my wallet(recommended)**, and you will find a page with a secret phrase (also known as **mnemonic phrase / seed phrase / master key**) containing 12 words mentioned in the picture below. This secret phrase is very important for **non-custodial wallets** like Metamask. You must copy and store this phrase in a safe place. If you lose it, you will not be able to retrieve your wallet again. Also, make sure that you store this secret phrase in a safe place because if someone gets this phrase he/she can access your account and do whatever they want.

5.1. Press **Reveal Secret Recovery Phrase**. Take a screenshot of that page.



6. On the next page, confirm your secret phrase; only then will the **confirm** button automatically be active. (You must type the phrases.) Press the confirm button to go to the next page.

1 2 3
Create password Secure wallet Confirm secret recovery phrase

Confirm Secret Recovery Phrase

Confirm Secret Recovery Phrase

1. love	2. ordinary	3.
4.	5. only	6. scene
7. gorilla	8.	9. term
10. bracket	11. you	12. bullet

Confirm

7. Our wallet has been created.



Wallet creation successful

You've successfully protected your wallet. Keep your Secret Recovery Phrase safe and secret -- it's your responsibility!

Remember:

- MetaMask can't recover your Secret Recovery Phrase.
- MetaMask will never ask you for your Secret Recovery Phrase.
- **Never share your Secret Recovery Phrase** with anyone or risk your funds being stolen
- [Learn more](#)

[Advanced configuration](#)

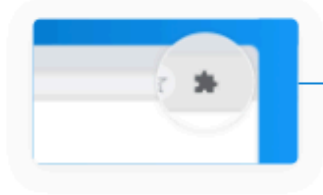
Got it!

8. Finally, pin the extension from the existing browser extension list, similarly mentioned in the picture below(Optional but recommended for this lab):

Your MetaMask install is complete!

Pin MetaMask on your browser so it's accessible and easy to view transaction confirmations.

1 Click the browser extension icon



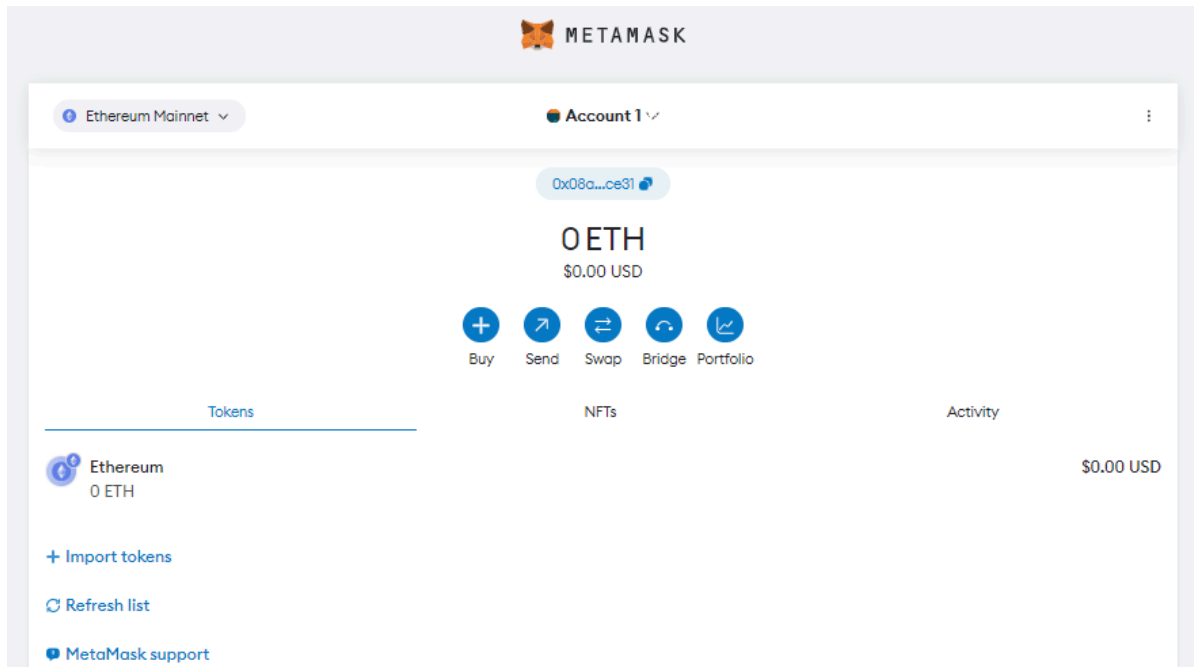
2 Pin MetaMask



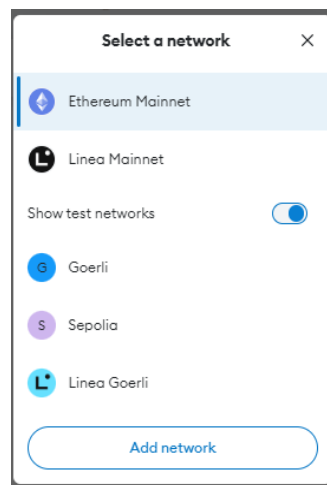
Next

Section-2 : Using the test network

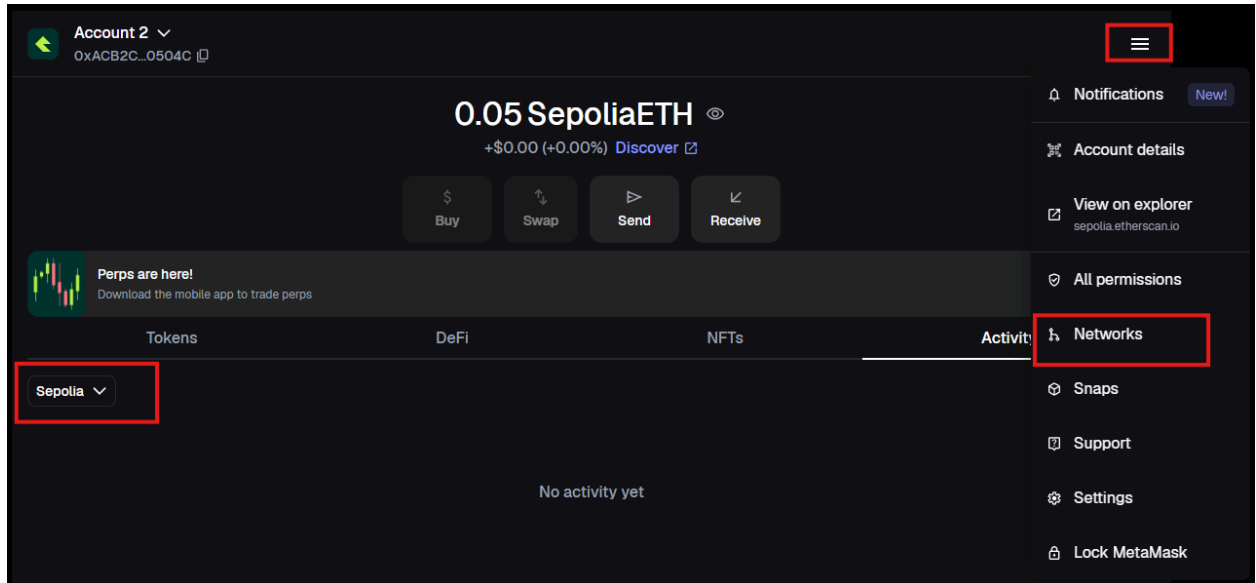
1. After successfully setting up the wallet, you will see a panel like the image below. Initially, you will have three networks to interact with, which are the Ethereum mainnet, Base Mainnet, and Linea. However, we will interact with a **test network**. Therefore, click the option “**Ethereum Mainnet**” with a dropdown option, which is on the left.



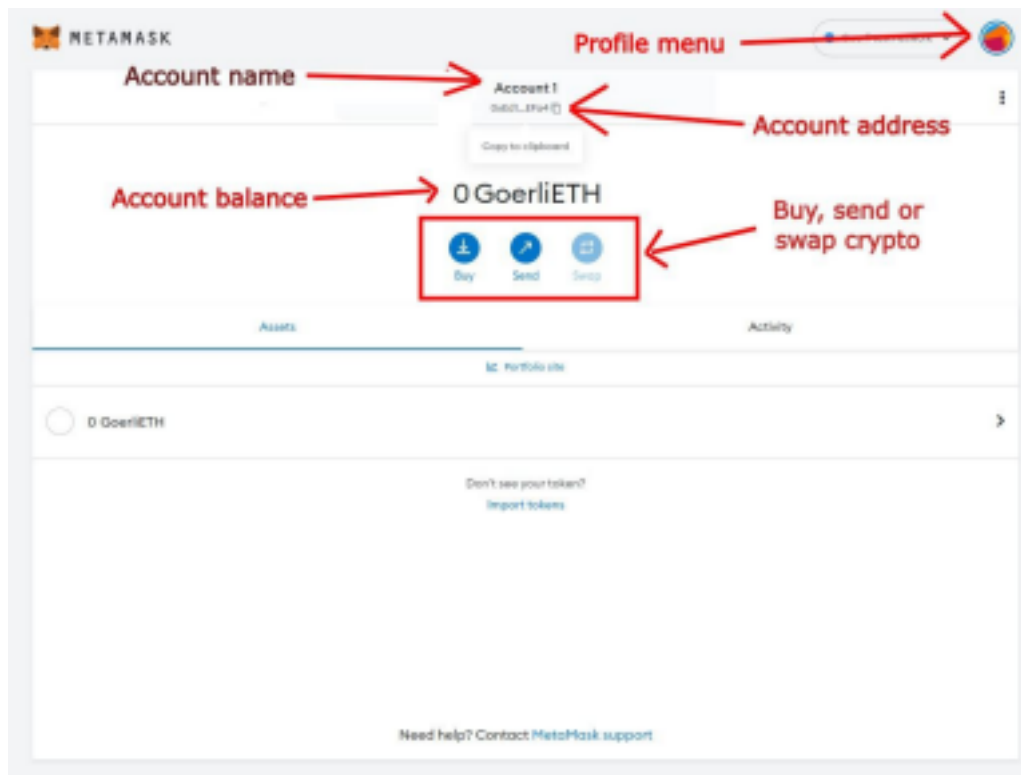
2. You will find a toggle button, and you should click “**show test networks**”. This will bring you some test networks in the wallet, like the image below. You can add more networks to the wallet. However, in this lab, we will try to interact with the Sepolia or Linea Goerli test network. First, click the “Sepolia” option.



In the newer Metamask version, this might not work, so click on the right hamburger icon and click on “**Networks**”. After that, you can follow the above instructions but if **Sepolia** isn’t getting selected, click edit and save to select it.



- These are the basic options of the Metamask wallet. We will interact with these options. However, the swap feature is not available in the test network; therefore, we will interact with other available options here.

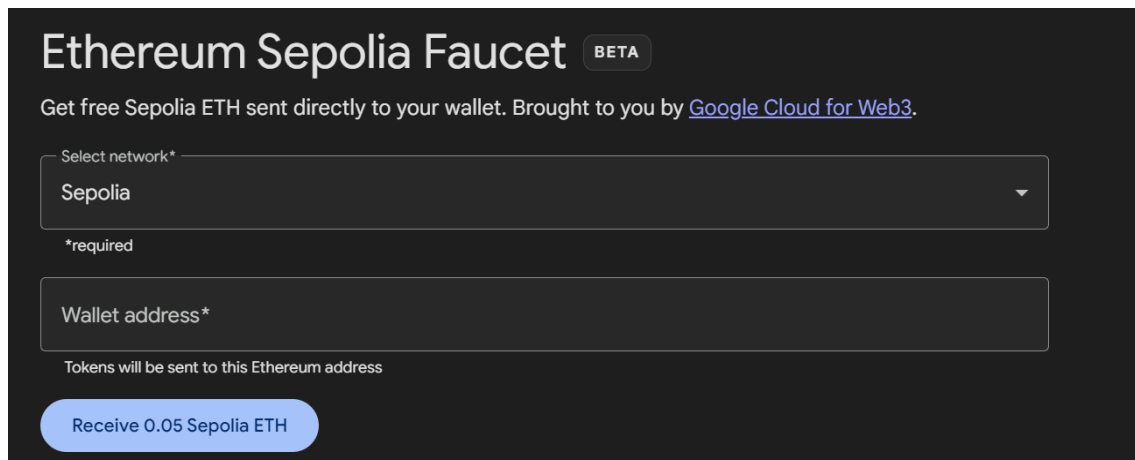


Section-3 : Interaction with the wallet.

In this section, we will fetch/buy some dummy cryptocurrency from the faucet and try to use the balance by sending it to another person/account.

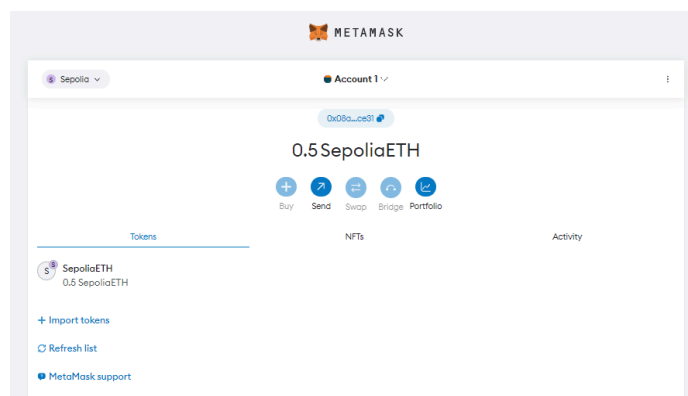
In the crypto world, faucets are applications or websites that provide users with a small amount of crypto, reward/currency/balance. In Ethereum, the currency they use is called ether or ETH. For Bitcoin, the currency name is Bitcoin or BTC. Similar to that, the Sepolia test network uses the currency called SepoliaETH. Now, let's get some balance/cryptocurrency, or to be more specific, SepoliaETH, since we are using the Sepolia test network.

1. Now visit <https://cloud.google.com/application/web3/faucet/ethereum/sepolia>



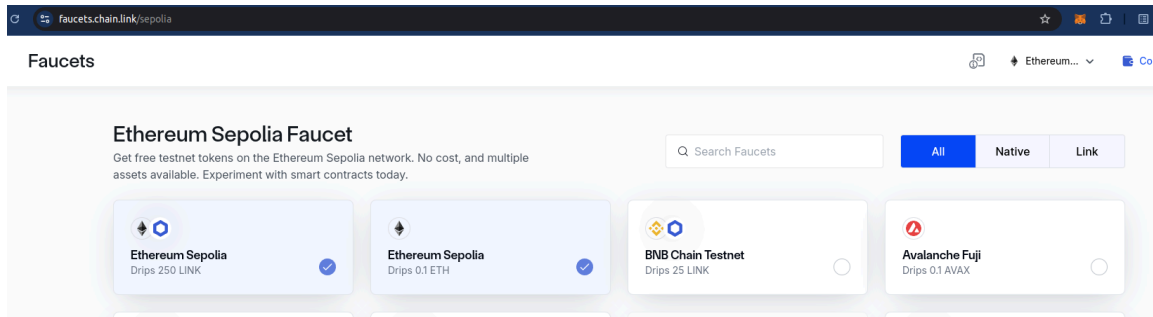
The screenshot shows the 'Ethereum Sepolia Faucet' interface. At the top, it says 'Ethereum Sepolia Faucet' with a 'BETA' badge. Below this, a message states: 'Get free Sepolia ETH sent directly to your wallet. Brought to you by Google Cloud for Web3.' There is a dropdown menu labeled 'Select network*' with 'Sepolia' selected. Below the dropdown is a text input field labeled 'Wallet address*'. A note below the input field says 'Tokens will be sent to this Ethereum address'. At the bottom, there is a blue button labeled 'Receive 0.05 Sepolia ETH'.

2. Paste your account address and press “Receive ...”
3. Now, visit Metamask you should be able to see 0.05 sepoliaETH in your account. Make sure you are in the **Sepolia** testnet. (It may take some time)



Another Faucet

- A. If the faucet before doesn't work, you can also try out this website:
<https://faucets.chain.link/sepolia> and **JUST CLICK CONTINUE**



- B. The prerequisite of this faucet is that you must have a **GitHub account**, and you need to authorize this faucet there to give some free rewards (Currently, most faucets require you to have actual ETH, and a few give you free rewards if you do some prerequisites).
- C. Now, click on **Continue with GitHub**, fill in your credentials, and authorize this faucet service. After that, it will bring you back to this page. Paste your wallet address in the two fields, then click on **Get Tokens**.

Add wallet addresses

Where are you sending the testnet tokens?

 Ethereum Sepolia Drips 0.1 ETH	→	Enter wallet address	 Remove
 Ethereum Sepolia Drips 250 LINK	→	Enter wallet address	 Remove

- D. Wait for a bit, and you should see the transactions be Successful.

Finished

Check to see if the tokens have arrived.

 Ethereum Sepolia Drips 0.1 ETH	✓	Transaction Hash 0x18bd...847e 	 Success
 Ethereum Sepolia Drips 250 LINK	✓	Transaction Hash 0xf787...22d2 	 Success

E. Go back to your wallet, and you should see 0.1 SepoliaETH was received.

ANOTHER FAUCET

- A. If the faucet above doesn't work, you can try out this faucet link:
<https://faucet.trade/sepolia-eth-faucet> (You have to have a Twitter account for this)
- B. Put your wallet address in the first box and, like Step 2 (before that, log in to your Twitter), follow @faucet_trade on Twitter and also tweet what they requested by clicking the 2nd link.

Available tokens

OG Newton Testnet [AOGI](#)

SaharaAI Testnet [SAH](#)

Soneium Minato Sepolia [ETH](#)

Holesky [ETH](#)

Morph Holesky [ETH](#)

Sepolia [ETH](#)

Berachain [BERA](#)

Tabi Chain V2 Carnival [TABI](#)

Blast Sepolia [ETH](#)

Polygon Amoy [MATIC](#)

BSC Testnet [BNB](#)

Avalanche Fuji [AVAX](#)

archived ▾

Buy testnet tokens
fast reliable deal without headache

Sepolia ETH Faucet

// obtain testnet tokens in just one minute

Step 1. Put your wallet address *

Step 2. Follow and tweet *

 Follow @faucet_trade

 Tweet

Step 3. Put your tweet URL *

Step 4. Solve captcha *

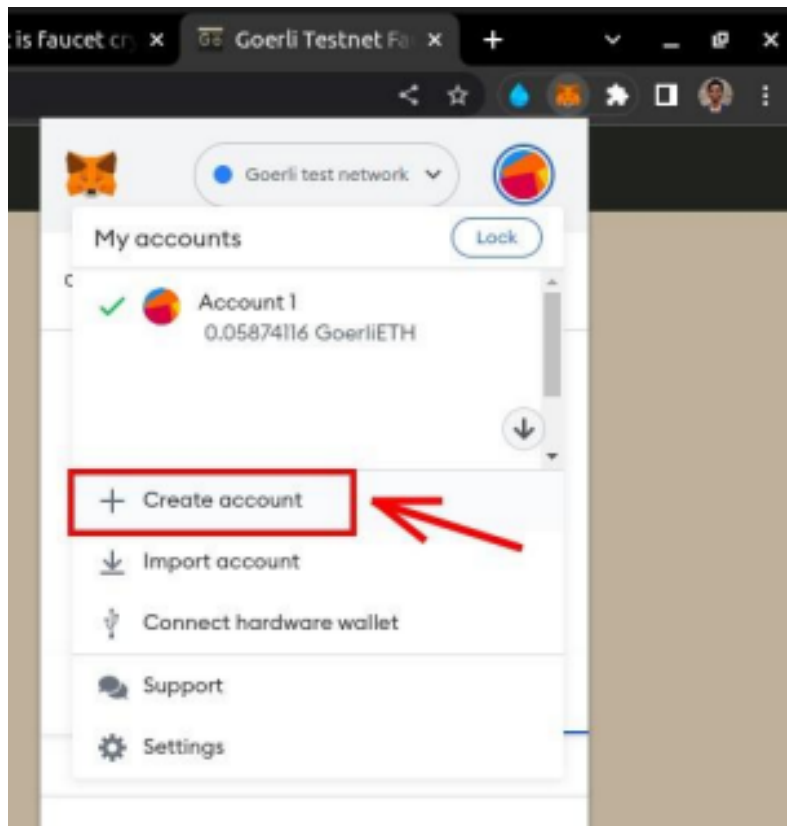
☐ I am human


Privacy - Terms

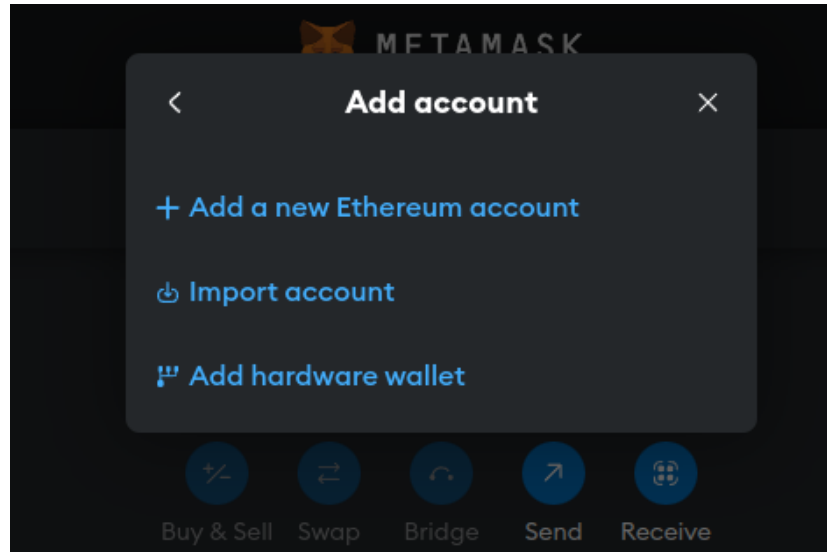
Receive tokens

- C. Now, take your Tweet link and put the URL in the Step 3 box.
- D. Finally, finishing the captcha, click on the **Receive tokens** button, and after a while, you will receive only 0.005 SepoliaETH.

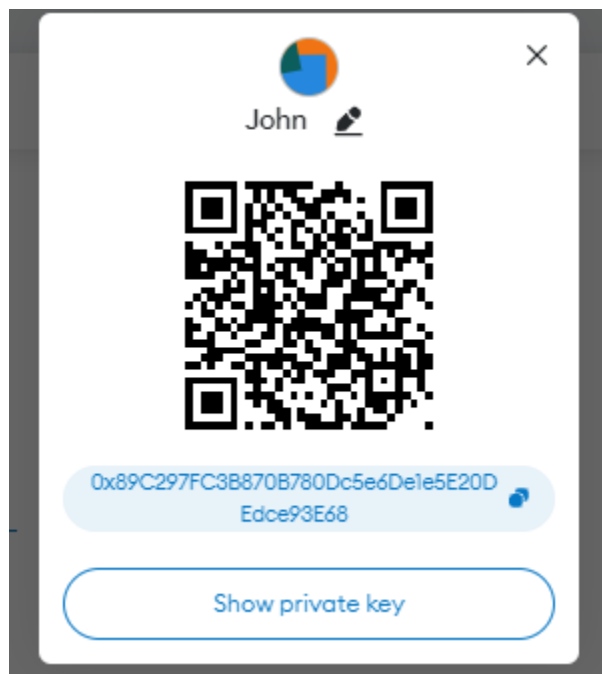
5. Now, we will send some cryptocurrency. But to whom ? Who is the receiver ? Users use their account address to send/receive the cryptocurrency in blockchain. This can be considered as their identity, like we use email addresses for sending/receiving email. Therefore, now we will create another account from the same wallet and send crypto from one account to another. You can create and manage multiple accounts from one wallet. Therefore, to create a new account, click the profile icon in Metamask and then click create new account. You can set the account name as you want. For this lab, I named it “**John**”.



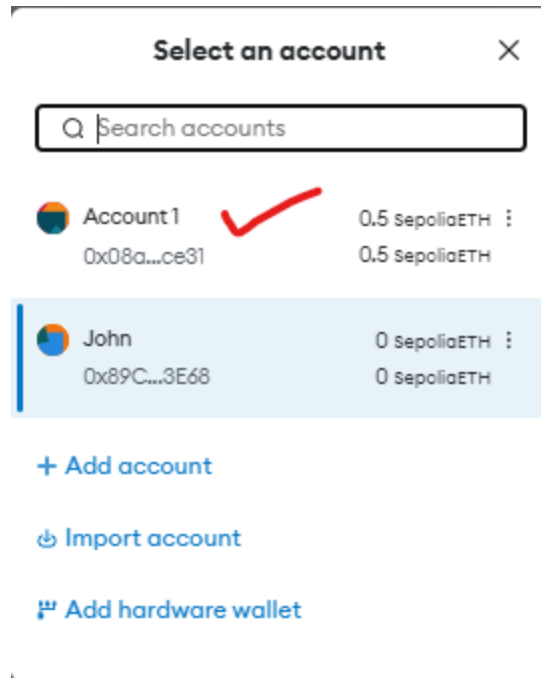
Or,



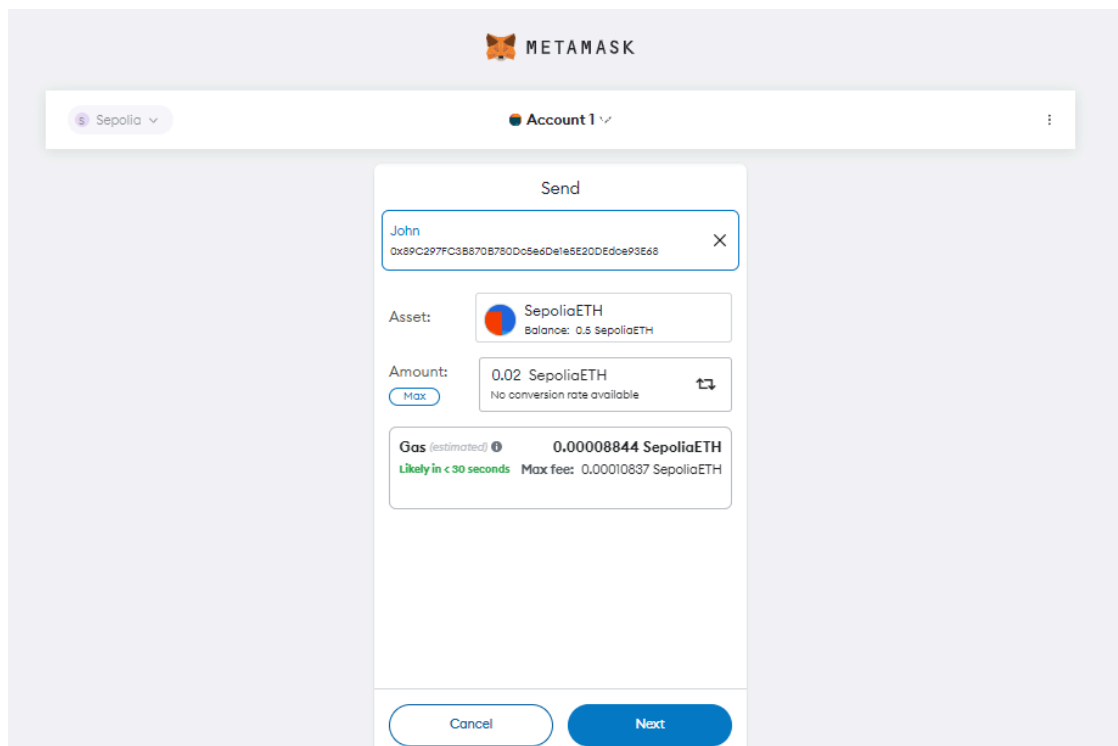
5. After creating a new account, you will see that the initial balance of that new account is zero. Now, we will send some crypto to this new account from Account 1. To do this, first copy the address of the newly created account.



6. Next, switch to Account 1 from the profile menu by clicking the account 1 option.

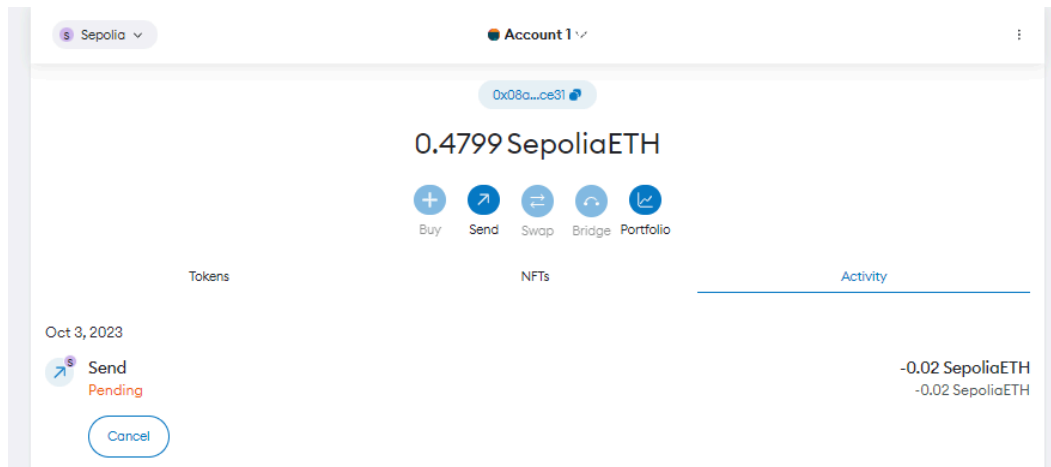


7. Now, click the **send** button, and then paste the account address, set an amount to transfer, and click next. Here, I am sending **0.02 SepoliaETH** to John (account 2).

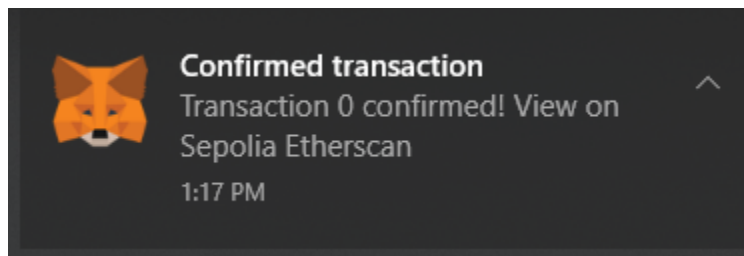


8. After clicking next, you will see a prompt to confirm transactions with related fees. This fee will be deducted from your account in addition to the **0.02 SepoliaETH** that you are going to send to John. In Ethereum, we pay some transaction fees(gas) for transactions.

After confirmation, you should see a pending transaction in the wallet.



9. After a few seconds, the transaction will be completed, and you will get a notification.



10. Now check the balance of both of your accounts. You should see Account 1 has a lower balance than before, and John (account 2) has a balance of **0.02 SepoliaETH**.

Checkpoint 1: Now, you have to send some amount of cryptocurrency to your friends. For this, get the account address of your group member/friend and then send some crypto to that account. After that, tell your friend/group member to send you some crypto. After completing it, show the result to your teacher.

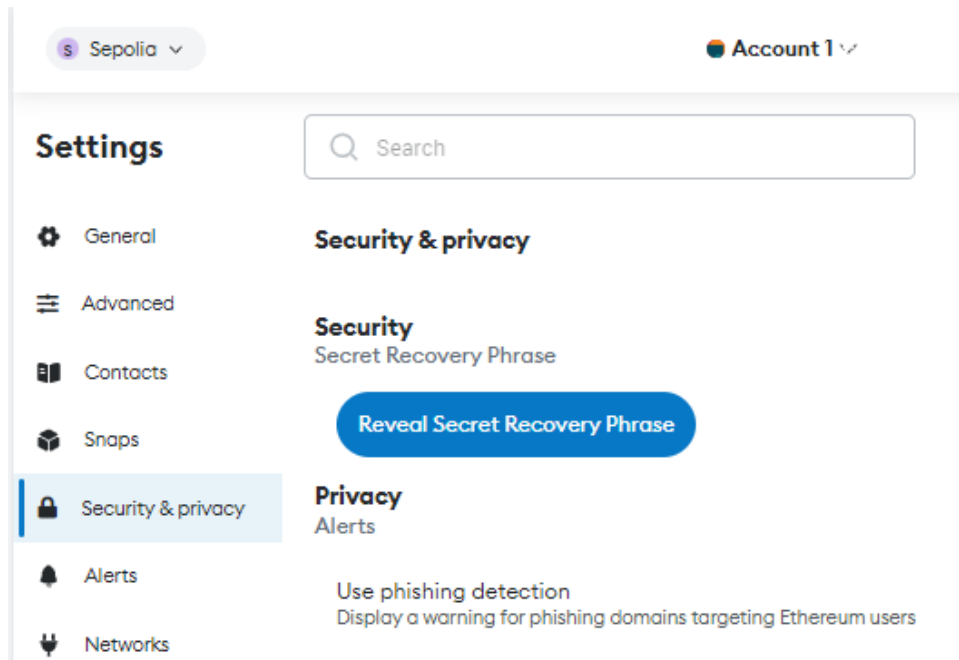
Section-4 : Backup and Retrieve Wallet and Accounts

In this section, we will discuss how we can back up our wallets and accounts so that we can access them in the future.

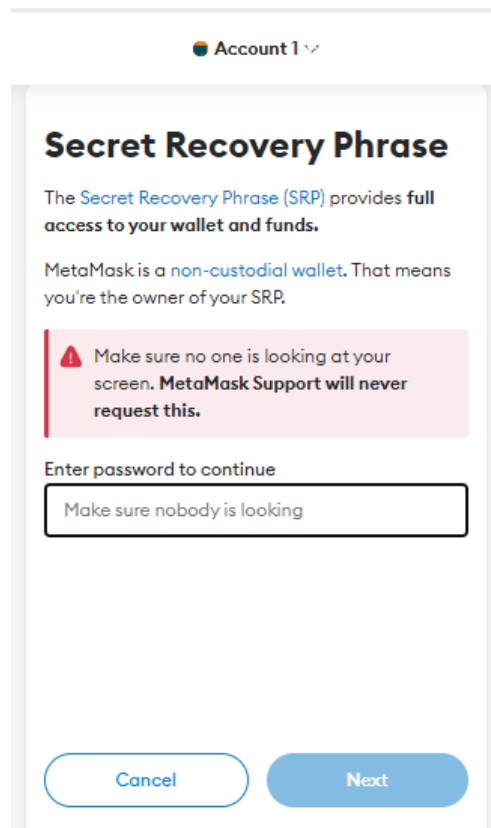
In Metamask, we have already seen the secret phrase. We know this secret phrase is essential to retrieve our wallet if we somehow lose the wallet. In addition, each account in a wallet contains a public key and a private key. In section 3, we sent and received some crypto using an address. This address is basically the public key of that respective account. Therefore, we will see how we can export these secret keys/phrases and use them in the future to retrieve the wallet/account.

4.1 Export Secret Phrase / seed phrase / master key of wallet

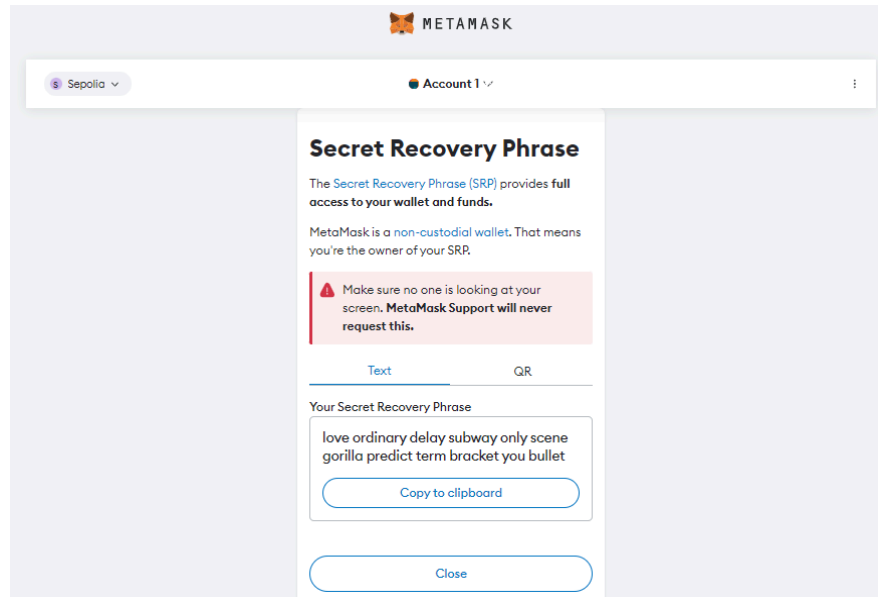
1. To export your wallet secret phrase, go to the “**settings > security and privacy**” under the Metamask profile menu. You will see an option to recover the secret phrase key.



2. Click it, and it will ask you two basic questions, which we already covered. Now answer them first. Then Metamask will prompt you to enter your password.



3. Once you enter your password, you will see your secret phrase like below:

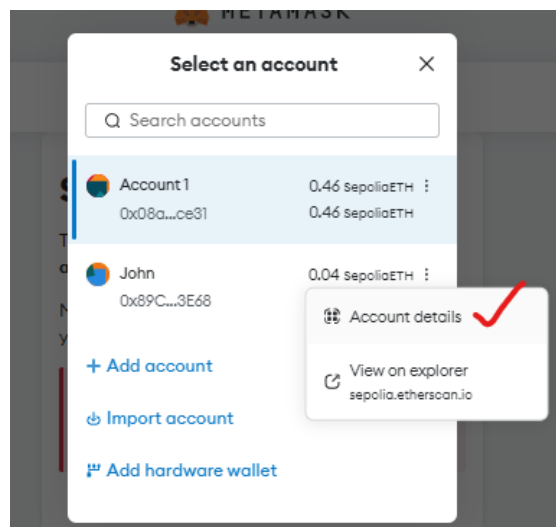


2. Click the “**Copy To Clipboard**” and paste the value, also if you want. Now you can store this secret phrase in a secure storage.

4.2 Export the private key of the account

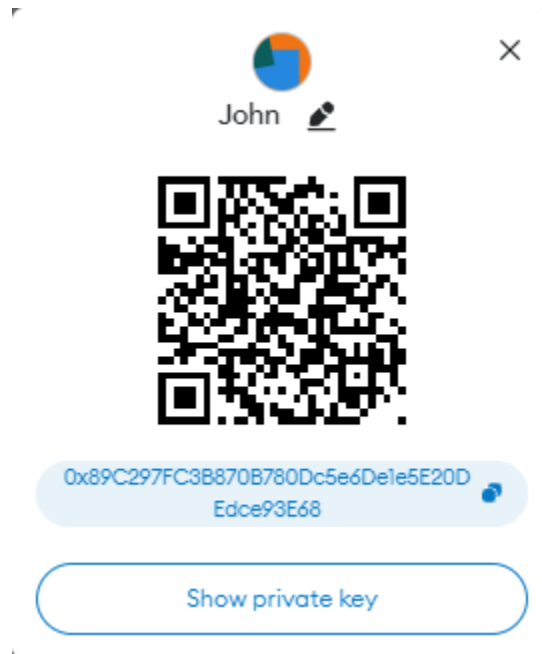
In a Metamask wallet, we can hold multiple accounts. You may want to back up your account so that you can access it from another wallet. To back up the account:

1. Open Metamask and click the menu button with three dots. This will bring up some options, and select “**Account Details**”



2. Then click “**Show private key**”. This will prompt you to enter your password. After

entering a password, you will see your private key for that respective account.



3. Copy this private key and store it in a safe place. If you lose this private key and also lose access to your wallet, you will not be able to retrieve this account anymore.

Checkpoint 2: Now, remove Metamask and try to install it again. But this time, try importing your wallet from the phrase that you stored in the previous section. Show this to your teacher.

Checkpoint 3: What did you see when you imported the wallet ? Only one account is there right ? What happens to others ? Are those deleted forever ? The answer is no. You can retrieve the accounts using their private key as we mentioned earlier. Therefore, now try to retrieve the account by importing instead of creating one from the Metamask. Give it a try. If you can do it successfully, please show this to your teacher.

Section-4 : Exploring transactions

In this section, you will learn about public blockchain explorers such as etherscan.io and bitaps.com. Your teacher will discuss these during the lab.